

- Q.6(a) The irreversible reaction $A \longrightarrow B$ was carried out in a batch reactor and the following concentration- time data were obtained 4M

t (min)	0	5	8	10	12	15	17.5	20
C_A (mol/dm ³)	4	2.25	1.45	1	0.65	0.25	0.06	0.008

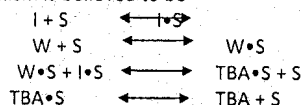
- (c) Determine the reaction order and specific reaction rate using differential method. 5M

Discuss the steps in a Langmuir-Hinshelwood kinetic mechanism.

Discuss dissociative adsorption with proper example and develop Langmuir isotherm for the same.

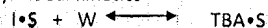
t-Butyl alcohol was produced by the liquid phase hydration (W) of isobutene (I) over an amberlyst-15 catalyst. The system is normally a multiphase mixture of hydrocarbon, water and solid catalysts. However, the use of co-solvents or the excess TBA can achieve reasonable miscibility.

The reaction mechanism is believed to be



Derive a rate law assuming;

- (i) The adsorption of isobutene is limiting.
(ii) The reaction follows Eley-Rideal kinetics



and that surface reaction is rate limiting.

Paper End

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Winter End Semester Examination: December 2019

Course: B. Tech. Chemical/ Petrochemical Engineering

Semester: V

Subject : Chemical Reaction Engineering-I [BTCH503]

Marks: 60

Date: 13/12/2019

Time: 03 Hrs.

INSTRUCTIONS TO THE STUDENTS

- Each question carries 12 marks.
- Attempt any five Questions in all.
- Illustrate your answers with neat sketches, diagrams etc. wherever necessary.
- Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that part is a part of examination.
- If some part of parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

Q.1

- (a) Give the informative account of different types of reactors used in industries with the design equations. 5M

- (b) Attempt ANY ONE of the following 3M

- (i) "Infinite reactor volume is necessary to reach complete conversion or equilibrium conversion", explain.
(ii) State true or false "the extent of reaction achieved in PFR does not depend on its shape". Justify your answer.

- (c) The first order reaction $A \rightarrow B$ is carried out in a tubular reactor in which the volumetric flow rate is constant. Derive an equation relating the reactor volume to the entering and exiting concentrations of A, the rate constant k, and the volumetric flow rate v. Determine the reactor volume necessary to reduce the exiting concentration to 10% of entering concentration when the volumetric flow rate is 10 dm³/min and specific reaction rate k is 0.23/min. 4M

Q.2 Attempt ANY THREE of the following 12

- (a) A second-order liquid phase reaction $A \rightarrow B$ is taking place in an idea mixed flow reactor (MFR). The exit conversion through MFR is 50 %. What will be the exit fractional conversion through a new MFR which is 6 times larger than the original reactor? M
- (b) A homogeneous gas phase decomposition reaction $4A \rightarrow B + 7S$ takes place in an isothermal ideal plug flow reactor. The reaction rate is, $-r_A = k_1 C_A$ with $k_1 = 0.17 \text{ s}^{-1}$; feed concentration of A (C_{A0}) = 0.1 mol/m³ Feed Flow rate (F_{A0}) = 0.17 mol/s. What is the size of the reactor to achieve 50% conversion?

Give the informative account of space velocity and space time with appropriate example.

(c)

A 400 liter CSTR and 100 liter PFR are available to process 1.0 liter of feed per second. The feed contains 41% A, 41% B and 18% inerts. The irreversible gas phase reaction, $A + B \rightarrow C$ is to be carried out at 10 atm and 227°C. The rate of reaction in g mol/lit min is given below as a function of conversion:

$-r_A$	0.2	0.0167	0.00488	0.00286	0.00204
X	0.0	0.1	0.4	0.7	0.9

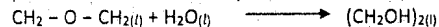
What would be the overall conversion if two 400 liter CSTRs were connected in parallel with half of the feed going to each reactor?

(d)

Give informative account of space velocity and space time with appropriate examples.

Q.3(a)

Set up a stoichiometric table for the following liquid phase reaction taking place inside a CSTR in the presence of H_2SO_4 (as a catalyst).



The initial concentrations of ethylene oxide and water are 1 lbmol/ft³ and 3.47 lbmol/ft³, respectively. If the conversion of ethylene oxide is 50% then what is the final concentration of ethylene oxide and water.

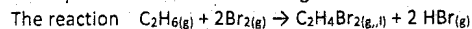
(b)

Give informative account of reaction order and molecularity.

(c)

(i)

Attempt ANY ONE of the following



is to be carried out at 200°C and 2500 kPa. The vapour pressure of 1,2-dibromoethane at 200°C is 506.5 kPa with $k=0.01 \text{ lit}^6/\text{mol}^2\text{min}$. The reaction is first order in ethane and second order in bromine. Calculate the conversion at which condensation begins.

(ii)

The rule of thumb that the rate of reaction doubles for a 10°C increase in temperature occurs only at a specific temperature for a given activation energy. Show that the relationship between activation energy and temperature for which rule holds is

Q.4

(a)

$$T = \left[\frac{10(K)E}{R \ln 2} \right]^{1/2}$$

Neglect any variation in concentration with temperature.

The elementary gas phase reaction



is carried out isothermally in a flow reactor with no pressure drop. The specific reaction rate at 50°C is 10^{-4} min^{-1} and $E=85 \text{ kJ/mol}$. Pure di-tert-butyl peroxide

enters the reactor at 10 atm and 127°C and a molar feed rate of 2.5 mol/min.

(b)

Calculate the reactor volume and space time to achieve 90% conversion in a

(i) CSTR

(ii) PFR.

6M

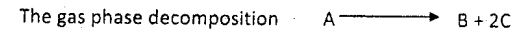
It is desired to design a CSTR to produce 200 million pounds of ethylene glycol per year by hydrolyzing ethylene oxide. However, before the design can be carried out, it is necessary to perform and analyze a batch reactor experiment to determine the specific reaction rate constant. Since the reaction will be carried out isothermally, the specific reaction rate will need to be determined only at the reaction temperature of the CSTR. At high temperatures there is significant by-product formation, while at temp. below 40°C the reaction does not proceed at a significant rate; consequently, a temp of 55°C has been chosen. Since the water is usually present in excess, its concentration may be considered constant during the course of the reaction. The reaction is 1st order in ethylene oxide.

In the lab experiment, 500 ml of a 2M solution of ethylene oxide in water was mixed with 500 ml of water containing 0.9 wt % sulfuric acid. The temperature was maintained at 55°C. The concentration of ethylene glycol was recorded as a function of time. From the table determine the specific reaction rate at 55°C.

Q.5
(a)

Time, min	0.0	0.5	1.0	1.5	2.0	3.0	4.0	6.0	10
Conc. of ethylene glycol (kmol/m ³)	0.0	0.145	0.270	0.376	0.467	0.610	0.715	0.848	0.957

6M



is carried out in a constant volume batch reactor. Runs 1 through 5 were carried out at 100°C, while run 6 was carried out at 110°C.

(i) From the following data, determine the reaction order and specific reaction rate.

(b)

6M

(ii) What is activation energy for this reaction?

Run	1	2	3	4	5	6
Initial Conc. C_{A0} (gmol/dm ³)	0.025	0.0133	0.01	0.05	0.075	0.025
Half-life, $t_{1/2}$ (min)	4.1	7.7	9.8	1.96	1.3	2.0

3M